

Performance Test

Test Run ID: c7b06caa-4643-431b-9689-b47f805197ff

This chapter provides a summary of the test run metrics. The tables contain the aggregated values of the metrics for the entire test run.

Date	05 June 2026	Max VUs Tested	80
Executor	constant-vus	Tool	k6

1. Executive Summary

The target endpoint for this performance test is the "https://quickpizza.grafana.com" URL. The testing type is a constant load scenario, where 80 Virtual Users (VUs) are maintained throughout the 45-second test duration. The load profile consists of a constant arrival rate of 80 VUs, with no ramping or stages, indicating a steady-state load test.

Metric	Value
Minimum TPS	39.50
Average TPS	43.87
Maximum TPS	137.16

2. Performance Summary

Trends

metric	avg	max	med	min	p90	p95	p99
http_req_duration	262.40ms	59555.19ms	234.95ms	214.59ms	246.56ms	256.46ms	-
http_req_waiting	261.95ms	59555.01ms	234.57ms	213.92ms	245.99ms	255.70ms	-
http_req_connecting	13.32ms	277.98ms	0.00ms	0.00ms	0.00ms	224.97ms	-
iteration_duration	1292.31ms	61006.08ms	1236.80ms	1215.55ms	1262.21ms	1688.07ms	-

Counters

metric	count	rate
data_received	9.77 MB	154.95 kB/s
data_sent	413.74 kB	6.41 kB/s
http_reqs	2,831	43.87/s
iterations	2,831	43.87/s

Rates

metric	rate
checks_succeeded	99.96% (2830/2831)
checks_failed	0.04% (1/2831)
http_req_failed	0.04% (1/2831)

Gauges

metric	value
vus	1 (min: 1, max: 80)
vus_max	80 (min: 80, max: 80)

Checks

Check Name	Passed	Failed	Success Rate	Status
status is 200	2830	1	99.96%	Fail

3. Advanced Analysis

3.1 Error Analysis

- Network/Connection errors:
 - Request timeout error: "Get \"https://quickpizza.grafana.com\": request timeout" (detected at 2026-06-05T13:16:36+05:30)
- Application errors:
 - Unknown fields in the options exported in the script (detected at 2026-06-05T13:15:33+05:30)

- **Infrastructure errors:** None detected

3.2 Observation

The observed failures and duration spikes can be attributed to the following reasons:

- The request timeout error indicates that the k6 test was unable to establish a connection with the target endpoint (<https://quickpizza.grafana.com>) within the specified timeout period. This could be due to network congestion, high latency, or issues with the endpoint itself.
- The unknown fields in the options exported in the script error is a warning message indicating that the k6 test encountered an unexpected field in the exported options. This is likely due to a configuration issue or a bug in the k6 script.
- The high duration spikes observed in the ``http_req_duration`` and ``iteration_duration`` metrics suggest that the k6 test was experiencing significant delays in processing requests. This could be due to a variety of factors, including high load on the target endpoint, slow database queries, or issues with the k6 test itself.

3.3 Final Recommendations

To resolve the identified bottlenecks and errors, the following recommendations are made:

1. **Advanced Connection Management:** Implement a connection pooling mechanism to reduce the overhead of establishing new connections to the target endpoint.
2. **Session Persistence sticky affinity/Redis cache:** Implement session persistence using sticky affinity or Redis caching to reduce the load on the target endpoint and improve response times.
3. **Gateway resilience Istio outlier circuit breaker:** Implement an Istio outlier circuit breaker to detect and prevent cascading failures in the gateway.
4. **Observability tracing and standardized log levels:** Implement observability tracing and standardized log levels to improve visibility into the k6 test and identify potential issues.
5. **Database read/write splitting and PgBouncer connection pooling:** Implement database read/write splitting and PgBouncer connection pooling to reduce the load on the database and improve response times.

3.4 Conclusion

The k6 performance test results indicate that the system experienced significant delays and failures during the test. While the test was able to complete successfully, the high duration spikes and request timeout errors suggest that the system may not be able to handle the expected load. To resolve these issues, the recommended architectural and database optimization changes should be implemented.